

QUESTION 1 (Start a new page)

- a) Solve the equation

$$4(x-2) = x-6$$

- b) Factorize completely

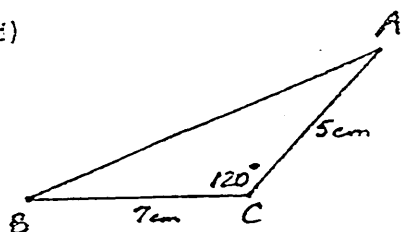
$$y^4 - 16$$

- c) Given that

$$R = \sqrt{\frac{S+P}{S-P}} \quad \text{and}$$

$S = 0.032$ and $P = 0.0235$, find R , correct to two decimal places.

- d)



In the given diagram,

$BC = 7 \text{ cm}$, $AC = 5 \text{ cm}$,

$\angle ACB = 120^\circ$

Find the exact length of AB .

- e) (i) Sketch the curve

$$(x-1)^2 + (y-2)^2 = 4$$

(ii) State whether or not it is a function.

(iii) Write down the domain.

QUESTION 2 (Start a new page)

- (a) Evaluate

$$\lim_{x \rightarrow 3} \frac{x^3 - 27}{x - 3}$$

- (b) In a container there are 3 red and 5 yellow discs. Three discs are chosen at random. Find the probability that:

- (1) all three will be red
- (2) all three will be the same colour

- (c) Given the points $A(-4, -6)$ and $B(2, 2)$ find:

- (i) the gradient of the line joining AB
- (ii) the midpoint M of AB
- (iii) the equation of the perpendicular bisector of the interval AB
- (iv) the perpendicular distance of the point $D(4, 0)$ from the perpendicular bisector found in part (iii)

QUESTION 3 (Start a new page)

(a) Differentiate (i) $3x^2 + \frac{3}{x^2}$

(ii) $x \log_e x$

(iii) e^{2-x}

(b) Find the exact value of:

$$\int_1^2 (5x - 2)^2 dx$$

(c) Find a primitive function of $x\sqrt{x}$

(d) For the parabola

$$y = (x - 4)^2 - 5,$$

write down: (i) the coordinates of its vertex

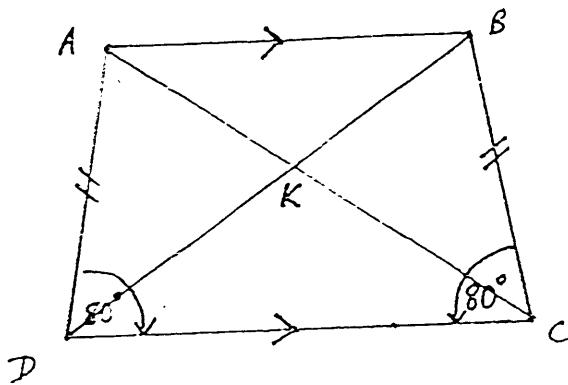
(ii) the minimum value of the function

(iii) the coordinates of its focus

QUESTION 4 (Start a new page)

- (a) The third term of an arithmetic series is (-4) and the tenth term is (-25) .
Find:
- (i) the first term
 - (ii) the common difference

(b)



In the diagram, ABCD is an isosceles trapezium with the angles ADC and BCD equal to 80° . The diagonals intersect at K.

- (i) Copy the sketch onto your answer paper
- (ii) Prove $\hat{ACD} = \hat{BDC}$
- (iii) Prove $\triangle ABK$ and $\triangle KDC$ are similar
- (iv) Name two isosceles triangles

QUESTION 5 (Start a new page)

- (a) Find the equation of the tangent to the curve

$$y = \frac{\log_e x}{x}$$

at the point (1, 0)

- (b) The function $f(x)$ is given by $f(x) = 3 + 5x + 2x^2 - 3x^3$
- (i) Find the coordinates of the turning points of $f(x)$ and determine whether they are maxima or minima.
- (ii) Draw a sketch of $y = f(x)$ in the domain $-\frac{3}{2} \leq x \leq \frac{3}{2}$
- (iii) Write down the maximum value of the function in this domain.

QUESTION 6 (Start a new page)

- (a) Graph on the number line the values of x for which

$$|4 - 2x| \geq 6$$

- (b) (i) Shade on a number plane the region R bounded by the curves

$$y = x^2 + 1 \quad \text{and} \quad y = 9 - x^2$$

- (ii) Find the area of this region.

- (c) Find the volume of the solid of revolution formed by rotating the portion of the curve $y = x^{\frac{2}{3}}$ from $0 \leq x \leq 1$ about the y - axis

QUESTION 7 (Start a new page)

- (a) If α and β are the roots of the quadratic equation

$$5x^2 - 4x + 2 = 0$$

evaluate

(i) $\alpha + \beta$

(ii) $\alpha\beta$

(iii) $(1 - \alpha)(1 - \beta)$

(iv) $\alpha^2 + \beta^2$

- (b) For what values of k does the quadratic equation

$$x^2 + 3 - k(x - 1) = 0$$

have equal roots?

- (c) Two students, John and Rebecca, attempt a problem in mathematics. The probability that John will solve the problem is $\frac{2}{3}$, whilst the probability that Rebecca will solve it is $\frac{3}{5}$.

- (i) What is the probability that John will solve it but Rebecca does not?

Another student, Sam, attempts the same problem. The probability that Sam will solve the problem is $\frac{7}{10}$.

- (ii) What is the probability that at least one of the three students will solve the problem?

QUESTION 8 (Start a new page)

- (a) Convert $0.\dot{3}8$ to a rational number in its simplest form.
- (b) A sinking ship S, which is 23 nautical miles from a rescue ship R, has a bearing of 034° from R. A lighthouse keeper K observes that S is on a bearing of 275° T from his position, *and R bears 255° T from his position.*
- (i) Draw a diagram marking on it the information supplied.
- (ii) Find the distance at of the ship S from the lighthouse at K. Give your answer to the nearest nautical mile.
- (c) Find the first term and the common difference of an arithmetic series in which

$$S_n = \frac{n^2}{3} + 5n$$

QUESTION 9 (Start a new page)

- (a) The minute hand of a watch is 8 mm long. How far does its tip move in 40 minutes?
- (b) The function $f(x)$ defined for x in the domain $3 \leq x \leq 7$ is given by the rule
- $$f(x) = \log_e (x^2), \quad 3 \leq x \leq 7$$
- (i) Draw up a table of values of $f(x)$ correct to one decimal place for
- $$x = 3, 4, 5, 6, 7$$
- (ii) Use this table to draw a sketch of the graph of $y = f(x)$, for $3 \leq x \leq 7$
- [Do not attempt to find turning points]
- (iii) Use Simpson's Rule with five function values to estimate the area between the curve, the x -axis, and the ordinates $x = 3$ and $x = 7$.
- (c) Find all real numbers x which satisfy the equation

$$x^4 = 2(4 - x^2)$$

Year 12 Trial 1990

Each question - 12 marks

Question 1.

a) $4(x-2) = x-6$

$4x-8 = x-6$

$3x = 2$

$x = \frac{2}{3}$

b) $y^2 - 16 = (y^2 + 4)(y^2 - 4)$
 $= (y^2 + 4)(y-2)(y+2)$

c) $R = \frac{\sqrt{0.032 + 0.0235}}{\sqrt{0.032 - 0.0235}}$

≈ 2.56

d) Using cosine rule,
 $AB^2 = 5^2 + 7^2 - 2 \cdot 5 \cdot 7 \cdot \cos 150^\circ$
 $= 25 + 49 - 70(-\frac{\sqrt{3}}{2})$
 $AB = \sqrt{74 + 35\sqrt{3}}$

2) A circle centre (1,2), radius 2 units

(i) $(x-1)^2 + (y-2)^2 = 0$

(ii) It is not a function

(iii) $\{x : -1 \leq x \leq 3\}$

Question 2.

a) $\lim_{x \rightarrow 3} \frac{x^2 - 17x + 30}{x^2 - 3x + 9}$
 $= \lim_{x \rightarrow 3} \frac{(x-3)(x-10)}{(x-3)(x+3)}$
 $= \lim_{x \rightarrow 3} \frac{x-10}{x+3}$
 $= \frac{9-10}{9+3} = -\frac{1}{12}$

b) $\pi(x) = 3; \pi(5) = 5; \pi(6) = 8$

(i) $P(R \cap R) = \frac{3}{6} \times \frac{2}{5} \times \frac{1}{6}$
 $= \frac{1}{60}$

(ii) $P(YYY) = \frac{5}{6} \times \frac{4}{6} \times \frac{3}{6}$
 $= \frac{1}{6}$

$P(\text{same colour}) = P(RRR) + P(YYY)$
 $= \frac{1}{60} + \frac{1}{6} = \frac{11}{60}$

c) (i) $m_{AB} = \frac{y_2 - y_1}{x_2 - x_1}$
 $A(-4, -6)$
 $B(2, 2)$
 $= \frac{2 - (-6)}{2 - (-4)} = \frac{8}{6} = \frac{4}{3}$

(ii) Midpoint is given by
 $(\frac{-4+2}{2}, \frac{-6+2}{2})$
 $= (-1, -2)$

Q2. (cont).

(iii) Line L is
 $\text{gradient of } L = -\frac{3}{4}$

Perpendicular bisector of
 AB through $(-1, -2)$ is given by
 $y+2 = -\frac{4}{3}(x+1)$
 $4y+8 = -4x-4$
 $3x+4y+11 = 0$

(iv) Perpendicular distance
 $= \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$
 $= \frac{|12 + 0 + 11|}{\sqrt{3^2 + 4^2}}$
 $= \frac{23}{5}$

Perp. distance = $\frac{23}{5}$ units

Question 3.

a) Let $y = 3x^2 + 3x - 2$
 $\frac{dy}{dx} = 6x - 6$
 $= 6x - \frac{6}{x^2}$

b) Let $y = x \log_e x$
 $\frac{dy}{dx} = \log_e x + 1$

(iii) $\frac{d}{dx}(e^{2-x}) = e^{2-x} \cdot -1$
 $= -e^{2-x}$

c) $\int_1^2 (5x-2)^2 dx = \left[\frac{(5x-2)^3}{15} \right]_1^2$
 $= \frac{8^3}{15} - \frac{3^3}{15}$
 $= 32\frac{2}{3}$

d) $\int x\sqrt{x} dx = \int x^{\frac{3}{2}} dx$
 $= \frac{2}{5} x^{\frac{5}{2}} + C$
 $= \frac{2}{5} x^2 \sqrt{x} + C$

e) $y = (x+4)^2 - 5$
 $(x+4)^2 = y+5$
 (i) Vertex $(-4, -5)$
 (ii) Minimum value is (-5)

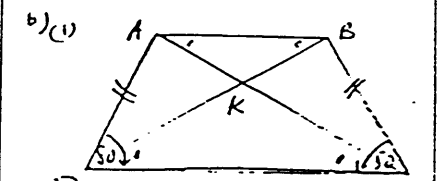
(iii) $a = \frac{1}{4}$
 Focus $(-4, -\frac{25}{4})$



MA556G (1)

Question 4.

a) $T_3 = -4$, $T_0 = -25$
 $a+2d = -4$ (1)
 $a+9d = -25$ (2)
 $(1) - (2) \Rightarrow -7d = 21$
 $d = -3$
 $a = 2$
 $\text{Sum of terms } = 2$
 $\text{Common difference} = -3$



Data: $ABCD$ is a rectangle
 $\angle ADC = \angle BDC = 90^\circ$
 (iii) Aim: To prove $\angle ADC = \angle BDC$
 Proof: In $\triangle ADC$, $\triangle BDC$
 $\angle ADC = \angle BDC$ (given)
 $AD = BC$ (given)
 DC is common
 $\therefore \triangle ADC \cong \triangle BDC$
 (two sides and included angle).

$\therefore \angle ACD = \angle BDC$
 (corresponding angles of congruent triangles)

(iii) Aim: To prove $AK \parallel KC$
 Proof: In $\triangle ABK$, $\triangle KDC$
 $\angle ABK = \angle KDC$ (alternate angles)
 $\angle KAB = \angle KCD$ (alt. angles)
 $\angle AKB = \angle DKC$ (vert. opp.)
 $\therefore \triangle ABK \cong \triangle KDC$ (ASA)
 $\therefore AK = KC$
 $\triangle KAC$ is isosceles triangle

Question 5.

a) $y = \frac{\log_e x}{x}$
 $\frac{dy}{dx} = \frac{x \cdot \frac{1}{x} - \log_e x \cdot 1}{x^2}$
 $= \frac{1 - \log_e x}{x^2}$

At the point $(1, 0)$
 $\frac{dy}{dx} = \frac{1 - \log_e 1}{1^2} = 1$

Equation of tangent
 is given by
 $y - 0 = 1(x - 1)$
 $y = x - 1$

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Question 5 (cont).

b) $f(x) = 3 + 5x + 2x^2 - 3x^3$

$f'(x) = 5 + 4x - 9x^2$

$f''(x) = 4 - 18x$

Turning points occur where $f'(x) = 0$.

$5 + 4x - 9x^2 = 0$

$(5 + 4x)(1 - x) = 0$

$x = 1, -\frac{5}{9}$

$f''(x) = 4 - 18x$

$f''(1) = 4 - 18$

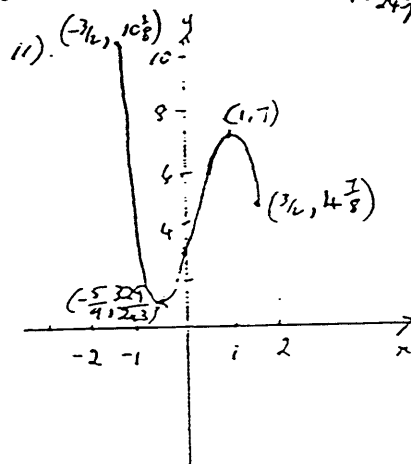
< 0

$\therefore$ maximum at $(1, 7)$

$f''(-\frac{5}{9}) = 4 + 10$

> 0

$\therefore$ minimum at $(-\frac{5}{9}, \frac{329}{27})$



When $x = 0$, $y = 3$

When $x = -\frac{3}{2}$, $y = 10\frac{1}{8}$

When $x = \frac{3}{2}$, $y = 4\frac{7}{8}$

Maximum value is $10\frac{1}{8}$

(7)

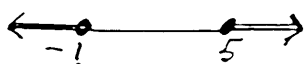
Question 6

a) $|4 - 2x| \geq 6$

$4 - 2x \geq 6$ $4 - 2x \leq -6$

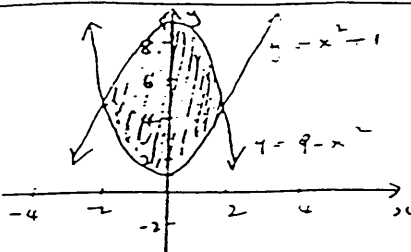
$-2x \geq 2$ $-2x \leq -10$

$x \leq -1$ $x \geq 5$



(3)

(4)



Points of intersection are given by

$x^2 + 1 = 9 - x^2$

$2x^2 = 8$

$x^2 = 4$

$x = \pm 2$

Required area is given by

$A = \int_{-2}^2 (9 - x^2) dx - \int_{-2}^2 (x^2 + 1) dx$

$= 2 \int_0^2 (8 - 2x^2) dx$

$= 2 \left[8x - \frac{2x^3}{3} \right]_0^2$

$= 2 \left[16 - \frac{16}{3} - 0 \right]$

$= \frac{64}{3}$

Area is $\frac{64}{3}$ sq. units

c) Required volume

is given by

$V = \pi \int_0^1 [f(y)]^2 dy$

$= \pi \int_0^1 y^3 dy$

$= \pi \left[\frac{y^4}{4} \right]_0^1$

$= \pi \left[\frac{1}{4} - 0 \right]$

$= \frac{\pi}{4}$

Volume is $\frac{\pi}{4}$ units³

(Working: $\frac{2}{3}$)

$y = x^2$

$y^3 = x^2$

When $x = 1$, $y = 0$

When $x = 1$, $y = 1$

(3)

Question 7.

a) $5x^2 - 4x + 2 = 0$

i) $\alpha + \beta = -\frac{b}{a}$

$= \frac{4}{5}$

ii) $\alpha\beta = \frac{c}{a}$

$= \frac{2}{5}$

(iii) $(1 - \alpha)(1 - \beta) = 1 - \beta - \alpha + \alpha\beta$

$= 1 - (\alpha + \beta) + \alpha\beta$

$= 1 - \frac{4}{5} + \frac{2}{5}$

$= \frac{3}{5}$

(iv) $x^2 + \beta^2 = (x + \beta)^2 - 2\alpha\beta$

$= \left(\frac{4}{5}\right)^2 - 2 \cdot \frac{2}{5}$

$= -\frac{4}{25}$

b) $x^2 + 3 - k(x - 1) = 0$

$x^2 - kx + (3 + k) = 0$

$\Delta = b^2 - 4ac$

$= k^2 - 4(3 + k)$

$= k^2 - 12 - 4k$

For equal roots, $\Delta = 0$

$k^2 - 4k - 12 = 0$

$(k - 6)(k + 2) = 0$

$k = -2, 6$

c) $P(D_s) = \frac{2}{5}$, $P(R_s) = \frac{3}{5}$

$P(S_s) = \frac{1}{10}$

(i) $P(D_s, R_f) = \frac{2}{3} \times \frac{2}{5}$

$= \frac{4}{15}$

(ii) $P(D_f, R_f, S_f) = \frac{1}{3} \times \frac{2}{5} \times \frac{3}{10}$

$= \frac{1}{25}$

iii) $P(\text{at least one solving it})$

$= 1 - \frac{1}{25}$

$= \frac{24}{25}$

(3)

Question 8

a) Let $x = 0.388 \dots$

$10x = 3.888 \dots$

$\therefore 9x = 3.5$

$x = \frac{3.5}{9}$

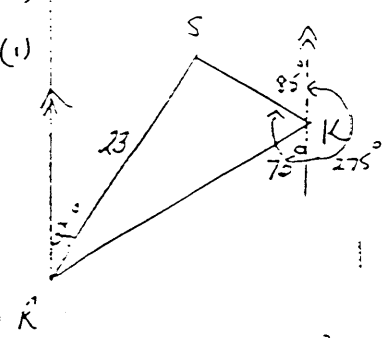
$= \frac{7}{18}$

(2)

(at least one)

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Question 9 (cont)



$255 = 180 + 75^\circ$

(iii) Sum of angles, $\angle SKR = 20^\circ$

$\angle SKR = 180 - (34 + 105) = 41^\circ$

(conclusion applies, with lines parallel)
In ΔSKR , using sine rule
 $\frac{SK}{\sin 41^\circ} = \frac{SR}{\sin 20^\circ}$
 $SK = \frac{23 \sin 41^\circ}{\sin 20^\circ} \approx 44$

Distance is approx. 44 m. (5)

(i) $S_n = \frac{n^2}{3} + 5n$

$T_n = S_n - S_{n-1}$
 $= \frac{n^2}{3} + 5n - \frac{(n-1)^2}{3} - 5(n-1)$
 $= \frac{n^2}{3} + 5n - \frac{n^2 - 2n + 1}{3} - 5n + 5$
 $= \frac{n^2 - n^2 + 2n - 1 + 15}{3}$
 $= \frac{2n + 14}{3}$

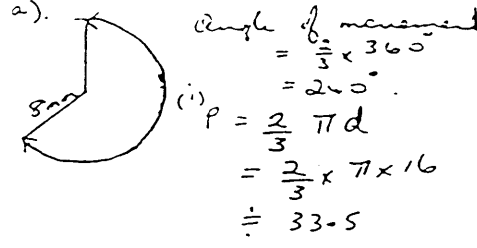
$T_1 = \frac{16}{3}$

$T_2 = \frac{18}{3}$

first 3 terms is $\frac{16}{3}$
common difference is $\frac{2}{3}$

$S_1 = \frac{1}{3} + 5 = \frac{16}{3}$ (1)
 $S_2 = \frac{4}{3} + 10 = \frac{34}{3}$ (2)
 $T_2 = S_2 - S_1$
 $= 6$ (1)
 $a = \frac{2}{3}$

Question 9

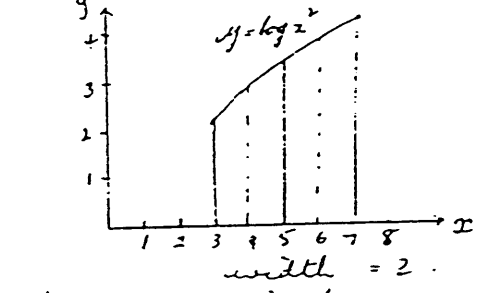


$C = \pi d$
angle of movement $= \frac{2}{3} \times 360^\circ = 240^\circ$
(i) $P = \frac{2}{3} \pi d$
 $= \frac{2}{3} \times \pi \times 16$
 ≈ 33.5
ce. (ii) $L = r\theta$
 $= 8 \times \frac{4\pi}{3}$
 ≈ 33.5 (2)

Length = 33.5 mm

(i) $f(x) = \log_e(x^2)$

x	3	4	5	6	7
$\ln x^2$	2.2	2.8	3.2	3.6	3.9



Area is given by
 $A = \frac{b-a}{6} \{f(a) + 4f(\frac{a+b}{2}) + f(b)\}$
Required area is given by
 $\approx \frac{2}{6} \{f(3) + 4f(4) + f(5)\}$
 $= \frac{1}{3} \{2.2 + 4 \times 2.8 + 3.2\}$
 $= 12.7$

Area is 12.7 units². (6)

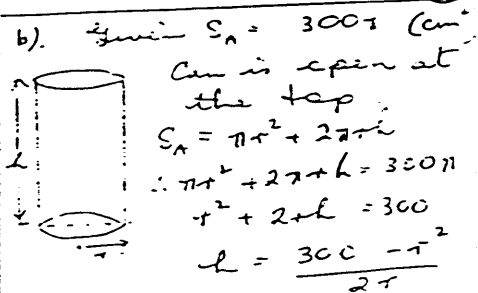
(c) $x^4 = 2(4 - x^2)$
 $x^4 + 2x^2 - 8 = 0$
Let $M = x^2$, equation becomes
 $M^2 + 2M - 8 = 0$
 $(M + 4)(M - 2) = 0$
 $M = -4$ or 2
 $x^2 = -4$ $x^2 = 2$
no solution $x = \pm\sqrt{2}$
Solution: $x = \pm\sqrt{2}$. (4)

Question 10

(i) $3^{10} + 3^9 + 3^8 + \dots$
series is geometric
 $a = 3^{10}$, $r = \frac{1}{3}$
 $S_\infty = \frac{a}{1-r}$
 $= \frac{3^{10}}{1-\frac{1}{3}}$
 $= \frac{3^{10}}{\frac{2}{3}}$
 $= \frac{3^{11}}{2}$

$S_{10} = \frac{a(1-r^{11})}{1-r}$
 $= \frac{3^{10}(1-(\frac{1}{3})^{11})}{1-\frac{1}{3}}$
 $= \frac{3}{2} \cdot 3^{10}(1-\frac{1}{3^{11}})$

(variation) $= \frac{3^{11}}{2} - \frac{3^{10}}{2}$
 $S_\infty - S_{10} = \frac{3^{11}}{2} - (\frac{3^{11}}{2} - \frac{3^{10}}{2})$
 $= \frac{3^{10}}{2}$ (5)



(i) $V = \pi r^2 h$
 $= \pi r^2 \cdot \frac{(300 - r^2)}{2r}$
 $= 150\pi r - \frac{\pi r^3}{2}$

(ii) $\frac{dV}{dr} = 150\pi - \frac{3\pi r^2}{2}$
 $\frac{d^2V}{dr^2} = -3\pi r$

for a maximum volume,
 $\frac{dV}{dr} = 0$ and $\frac{d^2V}{dr^2} < 0$
 $150\pi - \frac{3\pi r^2}{2} = 0$
 $\frac{3\pi r^2}{2} = 150\pi$
 $r^2 = 100$
 $r = \pm 10$

Since $r > 0$, $r = 10$
 $\frac{d^2V}{dr^2} = -30\pi$
 $\frac{d^2V}{dr^2} < 0$

$\therefore$ max. volume when $r = 10$
 $h = \frac{300 - 100}{20}$
 $= 10$

Maximum volume is given when radius is 10 cm
p. let $r = 10$.